MILITARY STANDARD

MIL-STD-1325-158
(NAVY)

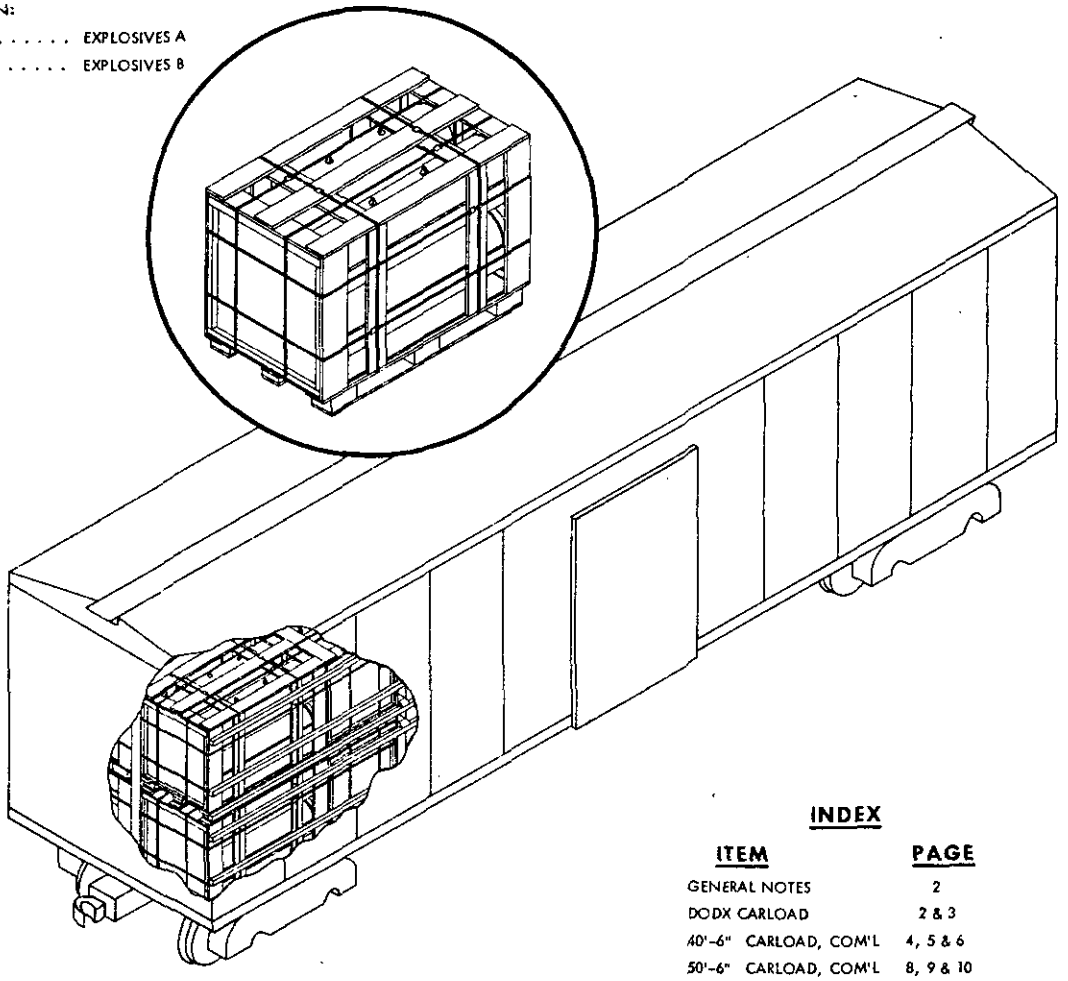
CARLOADING

9 JANUARY 1973

LAUNCHER, ROCKET, LAU-69/A
DOMESTIC UNIT LOAD

UNIT LOAD DATA

UNIT LOAD DRAWING WR-53/839
WEIGHT:
ROCKET . . . . . 1563 LBS
ROCKET MOTOR . . . . . 1163 LBS
DIMENSIONS . . . . . 65 3/4 L x 37 3/4 W x 44 7/8 H
CUBE . . . . . 64.5 CU FT
DOT HAZARD CLASSIFICATION:
ROCKET . . . . . EXPLOSIVES A
ROCKET MOTOR . . . . . EXPLOSIVES B



NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CONTAINERLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY)

INDEX

ITEM	PAGE
GENERAL NOTES	2
DODX CARLOAD	2 & 3
40'-6" CARLOAD, COM'L	4, 5 & 6
50'-6" CARLOAD, COM'L	8, 9 & 10

AUTHORIZED AND RELEASED FOR GENERAL USE

M.S. Gray NWHL 1/3/73
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J. E. Kelly 1/4/73
SIGNATURE (AIR) SYSCOM BY DIRECTION DATE

APPROVED BY BUREAU OF EXPLOSIVES

A. F. Grassmuck
DATE 12/20/72

ORIGINATOR Charles McBride 1/3/73
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
NAD EARLE, NEW JERSEY

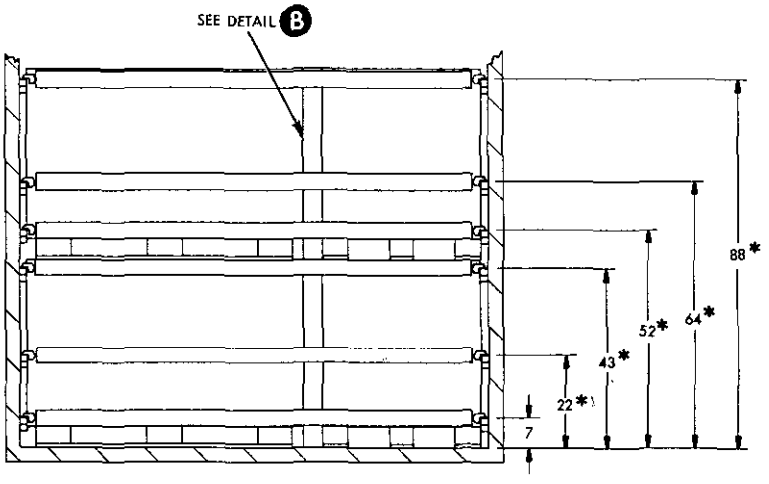
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GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN, AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) "RAILCAR LOADING OF HAZARDOUS MATERIALS."
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT. UTILITY LOADER CARS SHALL BE SERIES DODX 28000.
3. LOADING PLANS SHOWN ARE FOR DODX UTILITY LOADER CAR WITH 50 FT 6 INCHES INSIDE LENGTH, 107 3/4 INCHES INSIDE WIDTH BETWEEN RAILS (111 INCHES INSIDE WIDTH BETWEEN SIDE WALLS), COMMERCIAL BOXCARS WITH 40 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH, AND COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH.
4. THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.
5. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
6. *THE LOAD CONSISTS OF LAU-69/A LAUNCHERS PALLETIZED IN ACCORDANCE WITH WR-53/839 (DOMESTIC UNIT LOAD).*
7. THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORKLIFT TRUCK.
8. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY).
9. APPLICABLE MATERIAL SPECIFICATIONS:
   DUNNAGE LUMBER - FED. SPEC. MM-L-751
   NAILS - FED. SPEC. FF-N-105
10. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA, CLOSE AND SEAL BOXCAR DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF CAR AS PRESCRIBED IN OP 2165 (VOL. 1).

50 FT 6 IN. BOXCAR, DODX

1. THE CARLOAD CONSISTS OF 40 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. A DETAILED DESCRIPTION AND OPERATING INSTRUCTIONS FOR THE UTILITY LOADER ARE CONTAINED IN OP 1750.
3. TO PREVENT UNUSED "DF" EQUIPMENT FROM BECOMING DISLODGED DURING TRANSIT OF DODX CARS SECURE IT AT ANY LOCATION IN THE BOXCAR WHICH WILL NOT INTERFERE WITH UNLOADING.
4. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN DODX BOXCARS THE SAME PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE. ANY BAYS OR PORTION THEREOF MAY BE USED PROVIDING THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR ARE COMPLIED WITH (SEE MIL-STD-1325(NAVY)). EACH CROSSMEMBER WILL BE USED IN SUCH A MANNER THAT IT WILL RETAIN NOT MORE THAN 2200 LBS OF THE LADING.

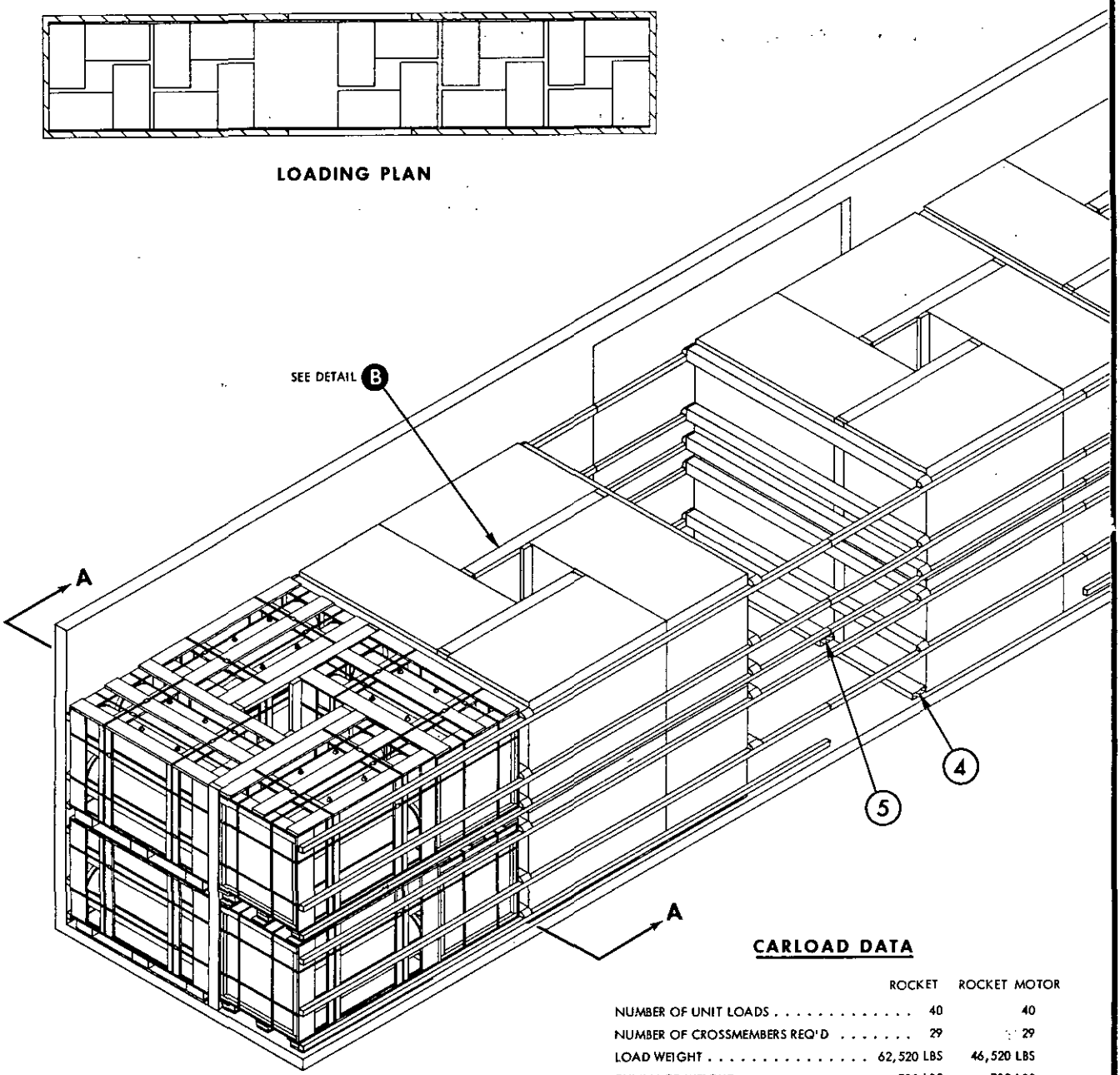


SECTION A-A

SHOWING LOCATION OF WALL MEMBERS
USED FOR CROSSMEMBERS
*DOORWAY MEMBER LOCATIONS

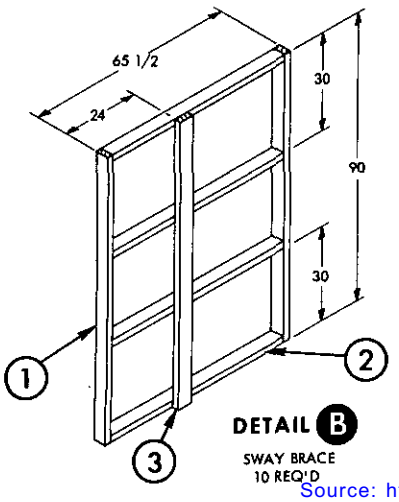
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LOADING PLAN



CARLOAD DATA

	ROCKET	ROCKET MOTOR
NUMBER OF UNIT LOADS	40	40
NUMBER OF CROSSMEMBERS REQ'D	29	29
LOAD WEIGHT	62,520 LBS	46,520 LBS
DUNNAGE WEIGHT	732 LBS	732 LBS
CARLOAD WEIGHT	63,252 LBS	47,252 LBS



DETAIL B
SWAY BRACE
10 REQ'D

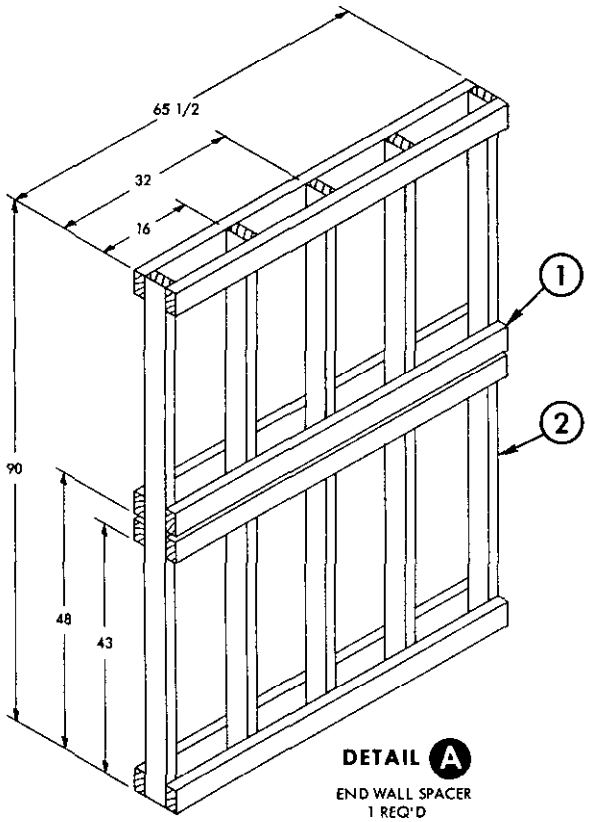
PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NAILS: NUMBER	NAILS: SIZE
5	LATERAL FLOOR MEMBER	2 x 6 x 65	1	CAR FLOOR	2 PER FOOT	16d
4	LATERAL FLOOR MEMBER	2 x 6 x 37	1	CAR FLOOR	2 PER FOOT	16d
3	SWAY BRACE SIDE VERTICAL	2 x 4 x 90	10	2	2 PER JOINT	10d
2	SWAY BRACE HORIZONTAL	2 x 4 x 62 1/2	40	SEE 1 & 3	-	-
1	SWAY BRACE END VERTICAL	2 x 4 x 90	20	2	2 PER JOINT	10d

LIST OF MATERIALS & NAILING DATA

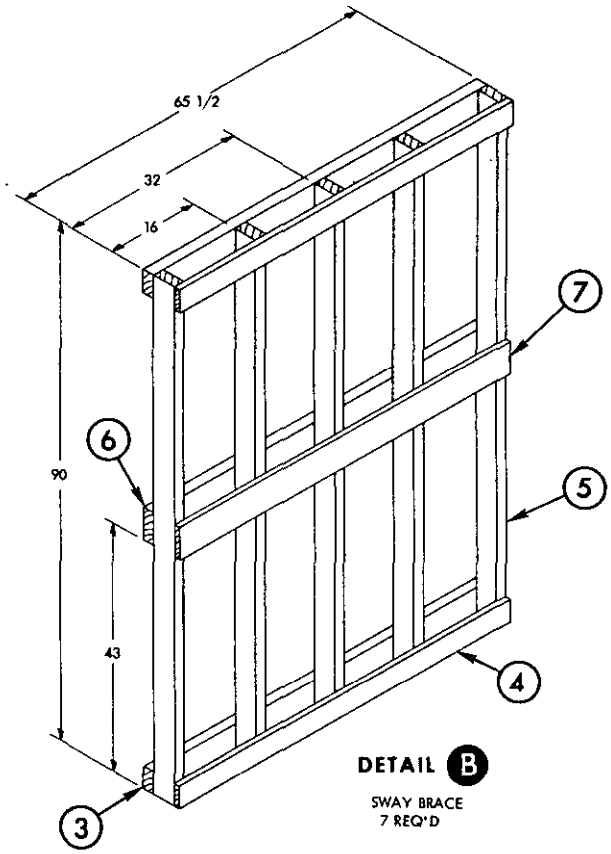
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40 FT 6 IN. BOXCAR, COMMERCIAL

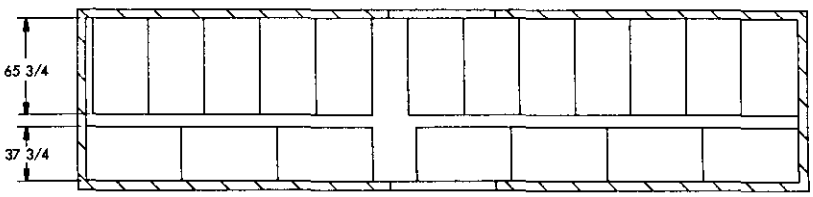
1. THE CARLOAD CONSISTS OF 38 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 6 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 6 FT WIDE.
4. WHEN BOXCARS OTHER THAN 9 FT 2 IN.WIDE ARE USED, THICKNESS OF PIECES 3, 4, 6, AND 7 OF DETAIL B MUST BE INCREASED OR DECREASED AS NECESSARY FOR A TIGHT LOAD ACROSS THE CAR.



DETAIL A
END WALL SPACER
1 REQ'D



DETAIL B
SWAY BRACE
7 REQ'D



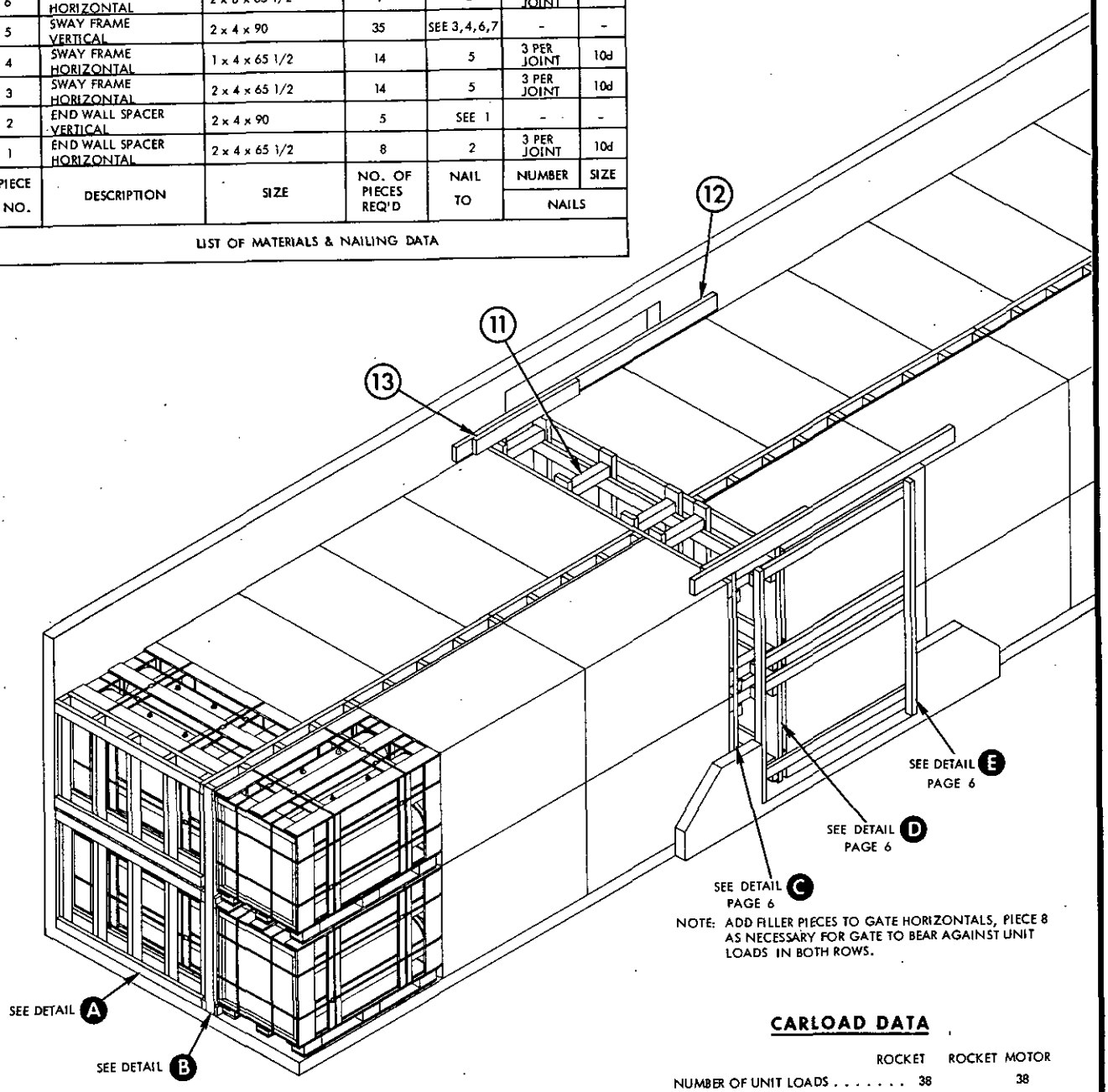
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PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE
15	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	8	14	2 PER JOINT	10d
14	VERTICAL DOORWAY MEMBER	2 x 3 x 92	4	DOOR POST	2 PER FOOT	20d
13	HOLDDOWN CLEAT	2 x 6 x 48	2	12	2 PER FOOT	10d
12	GATE HOLDDOWN	2 x 6 x 10 FT	2	CAR WALL	5 EA END	10d
11	STRUT	4 x 4 x WEDGE FIT*	20	9	2 PER JOINT	16d
10	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	9	3 PER JOINT	10d
9	CENTER GATE VERTICAL	2 x 6 x 92	10	SEE 8, 10	-	-
8	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	8	9	3 PER JOINT	10d
7	SWAY FRAME HORIZONTAL	1 x 6 x 65 1/2	7	5	3 PER JOINT	10d
6	SWAY FRAME HORIZONTAL	2 x 6 x 65 1/2	7	5	3 PER JOINT	10d
5	SWAY FRAME VERTICAL	2 x 4 x 90	35	SEE 3,4,6,7	-	-
4	SWAY FRAME HORIZONTAL	1 x 4 x 65 1/2	14	5	3 PER JOINT	10d
3	SWAY FRAME HORIZONTAL	2 x 4 x 65 1/2	14	5	3 PER JOINT	10d
2	END WALL SPACER VERTICAL	2 x 4 x 90	5	SEE 1	-	-
1	END WALL SPACER HORIZONTAL	2 x 4 x 65 1/2	8	2	3 PER JOINT	10d

LIST OF MATERIALS & NAILING DATA

* 2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4's.

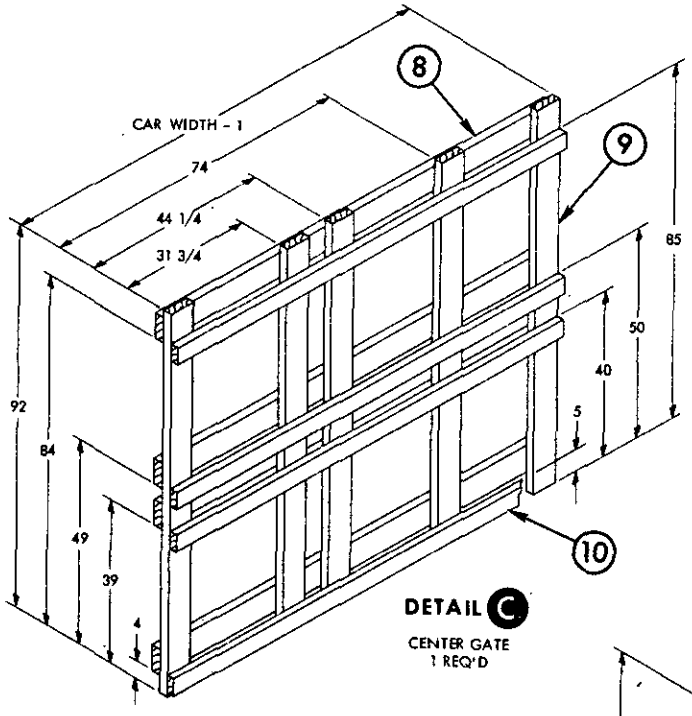


NOTE: ADD FILLER PIECES TO GATE HORIZONTALS, PIECE 8 AS NECESSARY FOR GATE TO BEAR AGAINST UNIT LOADS IN BOTH ROWS.

CARLOAD DATA

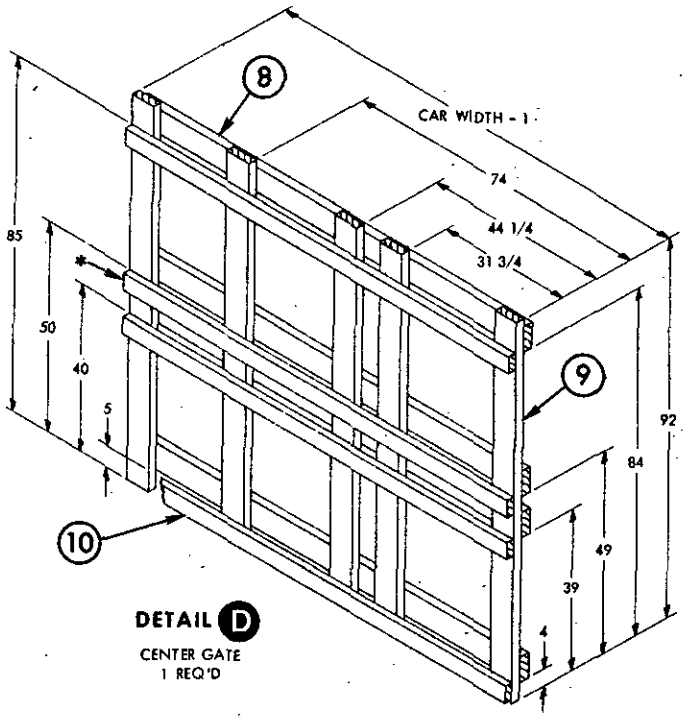
	ROCKET	ROCKET MOTOR
NUMBER OF UNIT LOADS	38	38
LOAD WEIGHT	59,394 LBS	44,194 LBS
DUNNAGE WEIGHT	1,680 LBS	1,680 LBS
CARLOAD WEIGHT	61,074 LBS	45,874 LBS

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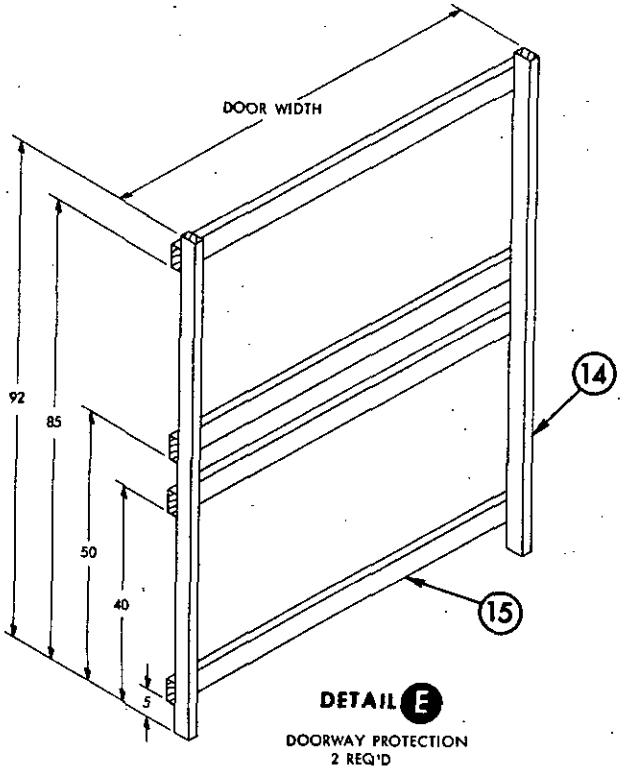


DETAIL C
CENTER GATE
1 REQ'D

* POSITION AFTER STRUTS HAVE BEEN INSTALLED ON LOWER STRUT CLEAT (REQ'D ONE GATE ONLY)



DETAIL D
CENTER GATE
1 REQ'D



DETAIL E
DOORWAY PROTECTION
2 REQ'D

WHEN BOXCAR HAS STEEL DOOR POSTS DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) PAGE 46, FIG. 21 IS REQUIRED.

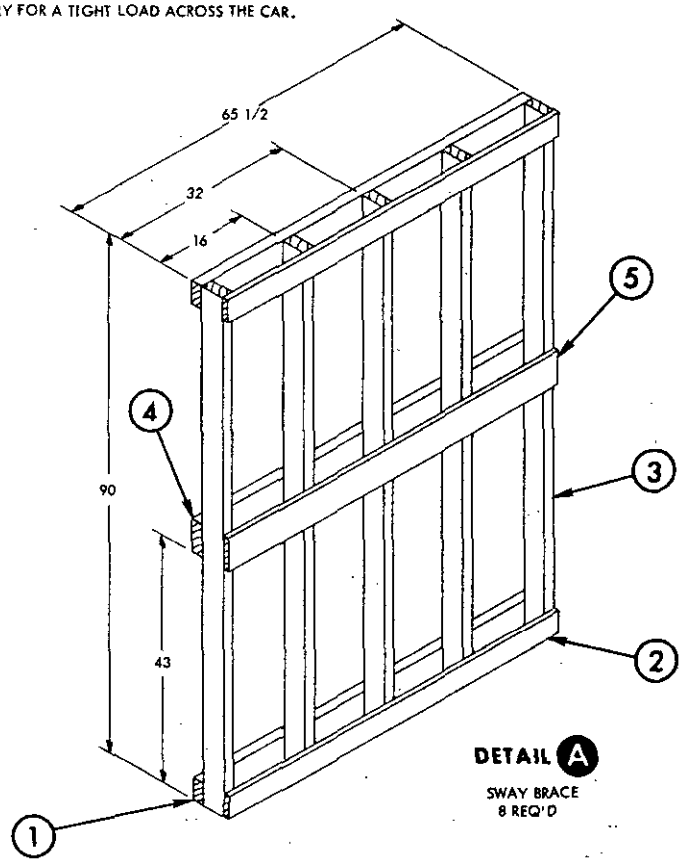
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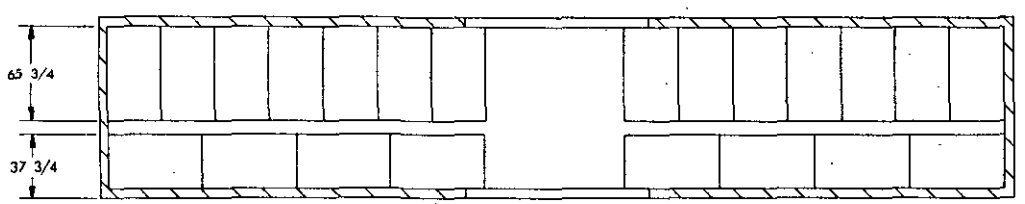
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50 FT 6 IN. BOXCAR , COMMERCIAL

1. THE CARLOAD CONSISTS OF 44 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/109. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE PROVIDING THE LENGTH OF PIECE 12 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS PROVIDING THE LENGTH OF PIECE 12 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR, AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE, RANDOM LENGTH MATERIAL, DOUBLED AND SPLICED, BUT WITH JOINTS OF SPLICES OFFSET, MAY BE USED. STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4" STEEL STRAPS PER STACK, EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS AND DOORWAY PROTECTION, PIECES 14 AND 15, IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED, EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.
5. WHEN BOXCARS OTHER THAN 9 FT 2 IN. WIDE ARE USED, THICKNESS OF PIECES 1, 2, 4 AND 5 OF DETAIL A MUST BE INCREASED OR DECREASED AS NECESSARY FOR A TIGHT LOAD ACROSS THE CAR.



DETAIL A
SWAY BRACE
8 REQ'D



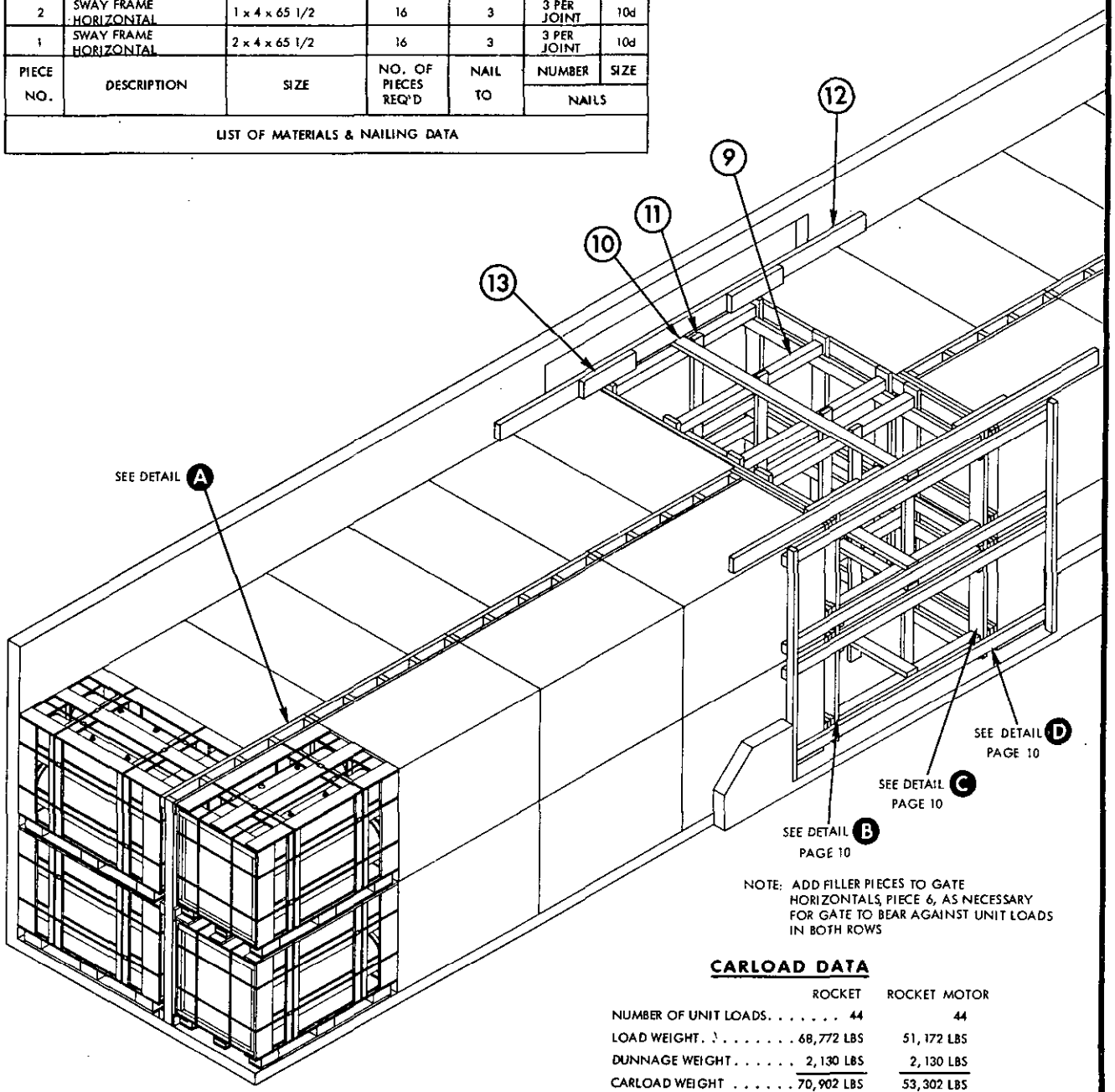
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PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER NAILS	SIZE
15	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	8	14	2 PER JOINT	10d
14	VERTICAL DOORWAY MEMBER	2 x 3 x 92	2	DOOR POST	2 PER FOOT	20d
13	HOLDDOWN CLEAT	2 x 6 x 24	4	12	5	10d
12	GATE HOLDDOWN	2 x 6 x 14 FT	2	CAR WALL	5 EACH END	10d
11	VERTICAL TIE BAR	2 x 4 x 92	5	9	2 PER JOINT	16d
10	HORIZONTAL TIE BAR	2 x 4 x CAR WIDTH-1	4	9	2 PER JOINT	16d
9	STRUT	4 x 4 x WEDGE FIT *	20	7	2 PER JOINT	16d
8	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	7	3 PER JOINT	10d
7	CENTER GATE VERTICAL	2 x 6 x 92	10	SEE 6, 8	-	-
6	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	8	7	3 PER JOINT	10d
5	SWAY FRAME HORIZONTAL	1 x 6 x 65 1/2	8	3	3 PER JOINT	10d
4	SWAY FRAME HORIZONTAL	2 x 6 x 65 1/2	8	3	3 PER JOINT	10d
3	SWAY FRAME VERTICAL	2 x 4 x 90	40	SEE 1, 2, 4, 5	-	-
2	SWAY FRAME HORIZONTAL	1 x 4 x 65 1/2	16	3	3 PER JOINT	10d
1	SWAY FRAME HORIZONTAL	2 x 4 x 65 1/2	16	3	3 PER JOINT	10d

LIST OF MATERIALS & NAILING DATA

*2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4's.

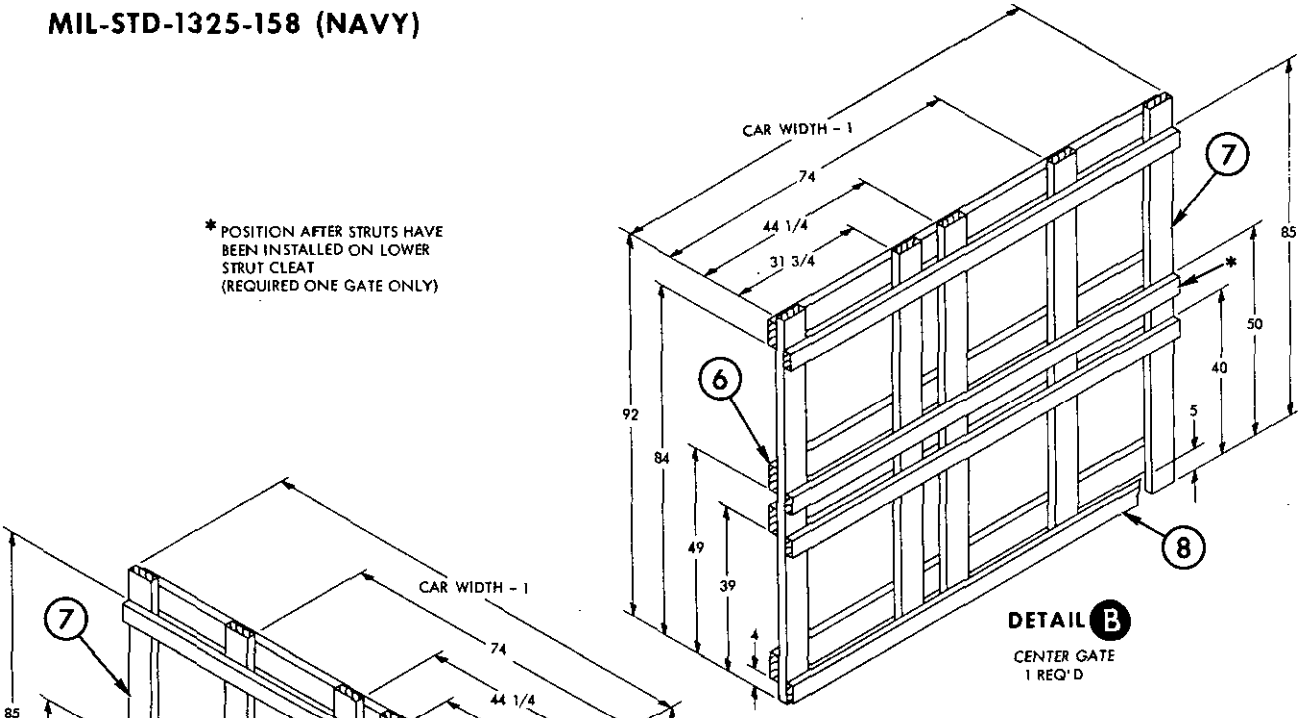


NOTE: ADD FILLER PIECES TO GATE HORIZONTALS, PIECE 6, AS NECESSARY FOR GATE TO BEAR AGAINST UNIT LOADS IN BOTH ROWS

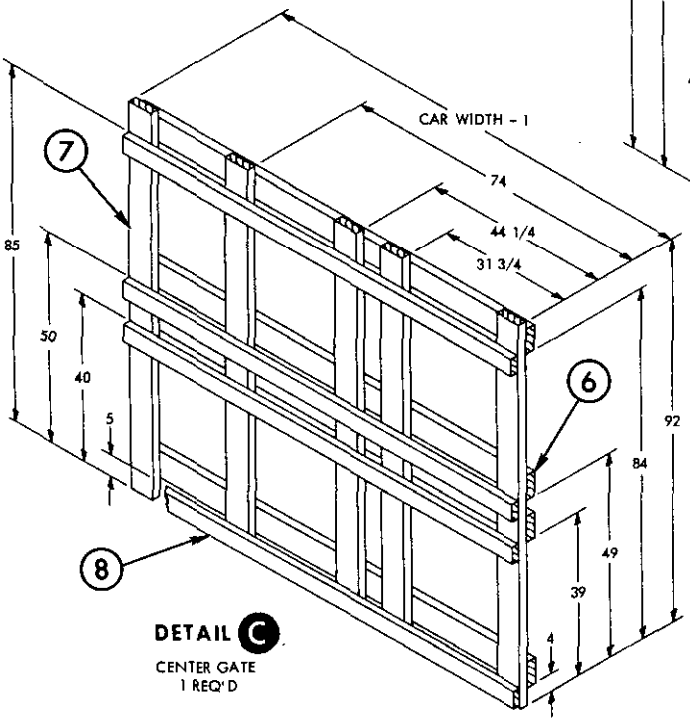
CARLOAD DATA

	ROCKET	ROCKET MOTOR
NUMBER OF UNIT LOADS	44	44
LOAD WEIGHT	68,772 LBS	51,172 LBS
DUNNAGE WEIGHT	2,130 LBS	2,130 LBS
CARLOAD WEIGHT	70,902 LBS	53,302 LBS

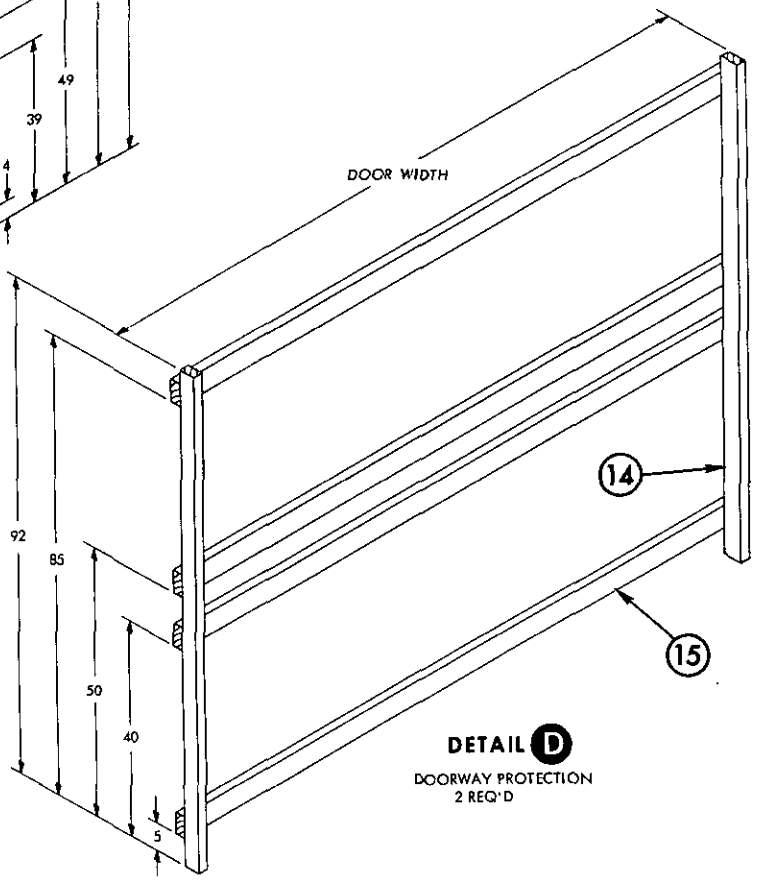
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DETAIL B
CENTER GATE
1 REQ'D



DETAIL C
CENTER GATE
1 REQ'D



DETAIL D
DOORWAY PROTECTION
2 REQ'D

WHEN BOXCAR HAS STEEL DOOR POSTS DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) *PAGE 46, FIG. 21 IS REQUIRED.*

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